



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Compound Measures (Speed, Density, Pressure)

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Your notes

Compound Measures

Compound Measures

What is a compound measure?

- A **compound measure** is something that is calculated by using **more than one** measurement
- Compound measures can be used to **measure rates**
 - This measures how much one quantity changes when the other is increased by 1
 - Examples include:
 - **Speed** – how much the distance changes for each unit of time
 - **Flow rate** – how much the volume changes for each unit of time
 - **Population density** – how many people there are for each unit of area
 - **Fuel consumption** – volume of fuel used for each unit of distance travelled

How do I find the units for a compound measure?

- You can use the **formula** for a compound measure to **derive its units**
 - Use the **units** for the **quantities** in the formula to derive the units of the compound measure
 - Write a **division** as **a/b** or **ab⁻¹** and pronounce it as “a per b”
- Examples include:
 - $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$
 - If the distance is measured in **km** and the time is measured in **minutes** then the speed is measured in **km/min** or **km min⁻¹**
 - $\text{Flow rate} = \frac{\text{Volume}}{\text{Time}}$
 - If the volume is measured in **m³** and the time is measured in **minutes** then the flow rate is measured in **m³/min** or **m³min⁻¹**

How do I find the formula for a compound measure?

- You can use the **units** for a compound measure to help remember its **formula**

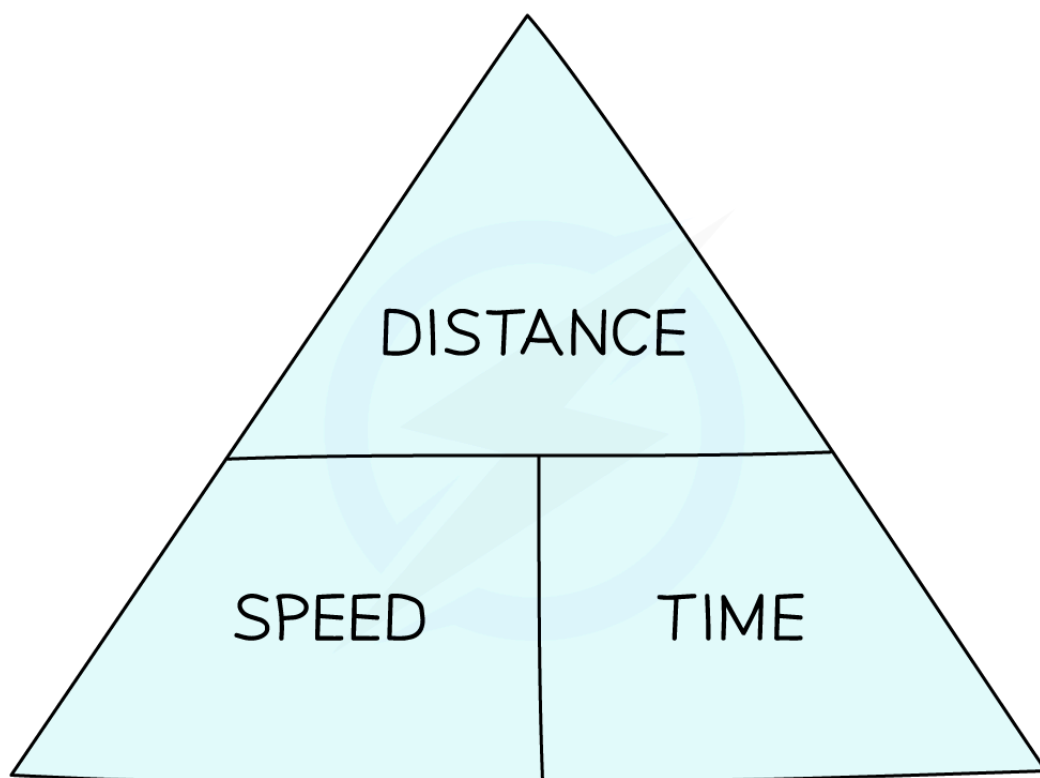


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- You just need to remember what each unit **measures**
- If the unit is **a/b** then the formula will be the quantity that **a** measures **divided** by the quantity that **b** measures
- Examples include:
 - **Density** can be measured in **kg/cm³**
 - kg is a measure of **mass** and cm³ is a measure of **volume**
- Therefore **Density** = $\frac{\text{Mass}}{\text{Volume}}$

What is a formula triangle?

- A **formula triangle** shows the **relationship** between the **different measures** in a **compound formula**
 - E.g. for Speed, Distance and Time



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- If you are calculating a variable on the **top of the triangle**, multiply the two variables on the bottom
 - For example, $\text{Distance} = \text{Speed} \times \text{Time}$
- If you are calculating a variable on the **bottom of the triangle**, divide the top by the other variable on the bottom
 - For example, $\text{Speed} = \text{Distance} \div \text{Time}$ and $\text{Time} = \text{Distance} \div \text{Speed}$



Examiner Tips and Tricks

- Check in the exam to see if the answer needs to be in different units
 - For example, the question may use metres and seconds but want the answer in km/h
- You need to remember the relationship between **speed**, **distance** and **time**



Worked Example

A high-speed racing car has an average fuel consumption of 3 km per litre during a race.

1 lap of the racing circuit is 5.9 km in length.

(a) Calculate the volume of fuel used, in litres, to complete 15 laps of the circuit.

The units for the fuel consumption are km per litre, which suggests the formula is

$$\text{fuel consumption} = \frac{\text{distance}}{\text{volume}}$$

Calculate the total distance covered for the 15 laps

$$15 \times 5.9 \text{ km} = 88.5 \text{ km}$$

Use the above formula to find the volume of fuel, V litres, used

$$\text{fuel consumption} = \frac{\text{distance}}{\text{volume}}$$

$$3 \text{ km per litre} = \frac{88.5 \text{ km}}{V \text{ litres}}$$

Rearrange the equation by multiplying both sides by V , and dividing both sides by 3



Your notes

$$3V = 88.5$$

$$V = \frac{88.5}{3} = 29.5$$

29.5 litres of fuel

The race car then requires a pit-stop to refuel to complete the final laps of the race.

The flow rate of the fuel pump is 720 litres per minute, and fuel is pumped into the car for 3.1 seconds.

(b) Calculate the volume of fuel, in litres, pumped into the car in this time.

The flow rate is 720 litres per minute which suggests the formula is $\text{flow rate} = \frac{\text{volume}}{\text{time}}$

Before we can use the formula, we need to change the units of time to both be the same

Change 720 litres per minute, into litres per second (to match the time fuel is pumped for, which is in seconds)

If 720 litres are pumped in 1 minute, 60 times less will be pumped in 1 second

$$720 \div 60 = 12 \text{ litres per second}$$

Substitute these values into the formula

$$12 \text{ litres per second} = \frac{\text{volume in litres}}{3.1 \text{ seconds}}$$

Multiply both sides by 3.1

$$\text{volume in litres} = 12 \times 3.1 = 37.2$$

37.2 litres

Speed, Density & Pressure



Your notes

Speed, Density & Pressure

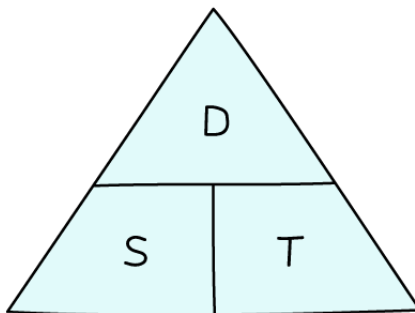
What are speed, density and pressure?

- **Speed, density and pressure** are frequently used **compound measures**
 - **Speed** is equal to **distance** divided by **time**
 - **Density** is equal to **mass** divided by **volume**
 - **Pressure** is equal to **force** divided by **area**

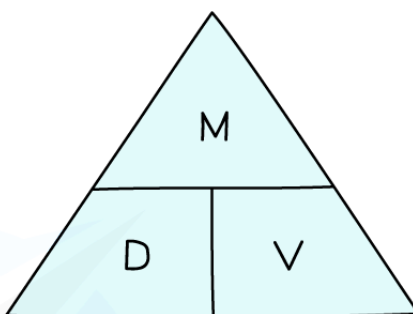


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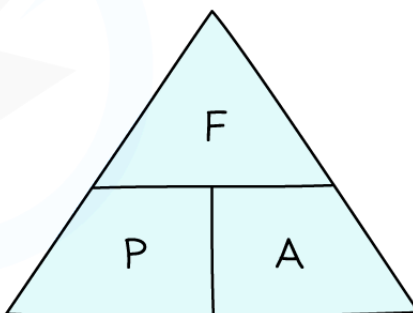
SPEED, DISTANCE, TIME



DENSITY, MASS, VOLUME

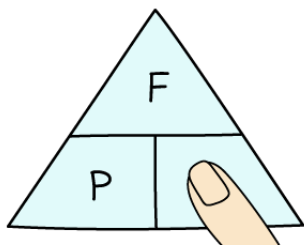


PRESSURE, FORCE, AREA



BLUE TRIANGLES WORK BY COVERING UP THE MEASURE YOU ARE TRYING TO FIND

e.g. FOR FINDING AREA...



$$\text{AREA} = \frac{\text{FORCE}}{\text{PRESSURE}}$$



Your notes

What should I know about speed, distance and time?

- **Speed** is commonly measured in **metres per second (m/s)** or **kilometres per hour (km/h)**
 - The units indicate **speed** is **distance per time**

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

- '**Speed**' (in this formula) means '**average speed**'
- In harder problems there are often **two journeys** or **two parts to one longer journey**

What should I know about density, mass and volume?

- Density is usually measured in **grams per centimetre cubed (g/cm³)** or **kilograms per metre cubed (kg/m³)**
 - The units indicate that **density** is **mass per volume**

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

- You may need to use a **volume formula** to find the volume of an object first

What should I know about pressure, force and area?

- Pressure is usually measured in Newtons per square metre (**N/m²**)
 - The units of pressure are often called **Pascals (Pa)** rather than **N/m²**
 - The units indicate that **pressure** is **force per area**

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

- Remember that **weight** is a **force**
 - It is different to **mass**



Examiner Tips and Tricks

- Look out for a mixture of units
 - **Time** can be given as **minutes** but common phrases like '**half an hour**' (30 minutes) could also be used in the same question

- Any mixed units should be those in common use and easy to convert, e.g., **g** to **kg** or **m** to **km** etc



Worked Example

A box exerts a force of 140 newtons on a table.

The pressure on the table is 35 newtons/m².

$$p = \frac{F}{A}$$

p = pressure
 F = force
 A = area

Calculate the area of the box that is in contact with the table.

Method 1

Substitute the numbers you know into the formula

$$35 = \frac{140}{A}$$

Solve the equation for A

First multiply both sides by A to get A out of the denominator

$$35A = 140$$

Then divide by 35 to find the value of A

$$A = \frac{140}{35}$$

The units will be m², matching the units seen in newtons/m²

$$4 \text{ m}^2$$

Method 2

Use the given formula to create a formula triangle for pressure, force and area

$$\begin{array}{ccc} & F & \\ p & & A \end{array}$$

A is on the bottom of the triangle, so this tells us to divide F by P

$$A = \frac{F}{p} = \frac{140}{35} = 4$$

 4 m^2 

Your notes



Worked Example

The density of pure gold is 19.3 g/cm^3 .

What is the volume of a gold bar that has a mass of 0.454 kg ?

Begin by checking that all of the units are consistent

Density is given in g/cm^3

Convert the volume of the gold bar into grams to match the units

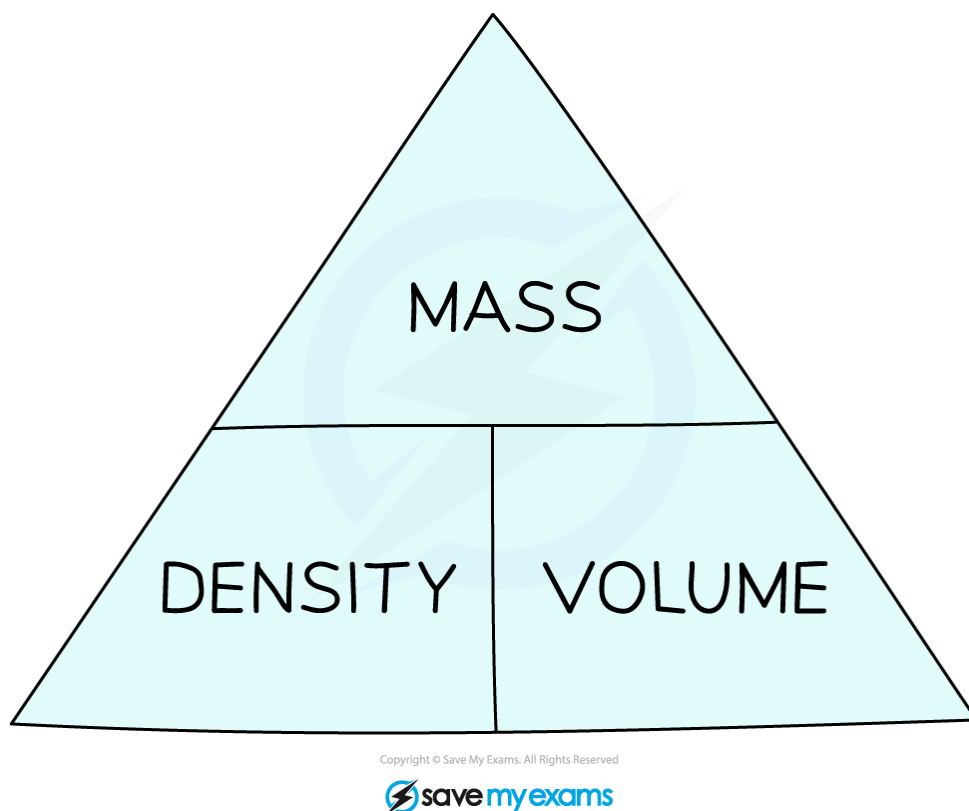
$$0.454 \text{ kg} = (0.454 \times 1000) \text{ g} = 454 \text{ g}$$

The units of density are g/cm^3 , so divide the mass (g) by the volume (cm^3)

Or, write out the formula triangle



Your notes



Write out the formula that you will need

$$\text{Volume} = \frac{\text{mass}}{\text{density}}$$

Substitute the given values for the mass and the density

$$\text{Volume} = \frac{454}{19.3} = 23.5233\dots$$

Make sure you give the correct units with your final answer

The density is given in g/cm^3 , so the volume should be cm^3

$$\text{Volume} = 23.5 \text{ cm}^3 \text{ (1 d.p.)}$$